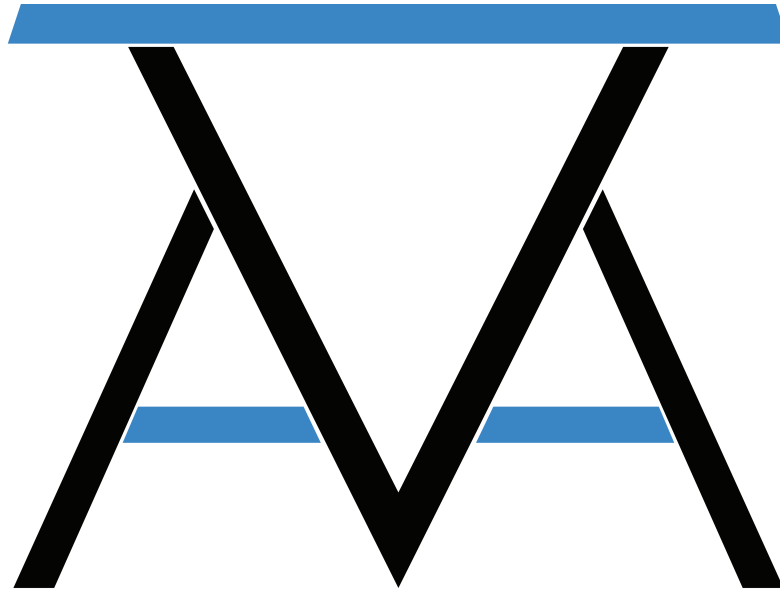




ADVANCED VTOL AIRCRAFT

Group Design Project

Individual Report - FATT04 Systems and Propulsion

Student: Jorge Van Looy Lopez
CID: 01101738

Department: Department of Aeronautics
Module: AE3-306 Group Design Project
Primary Supervisor: Dr. M. Ribera Vicent
Secondary Supervisor: Dr. Z. Sharif Khodaei
Academic Year: 2017/2018

Personal Tutor: Professor E. Greenhalgh
Date: June 18, 2018

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.

Abstract

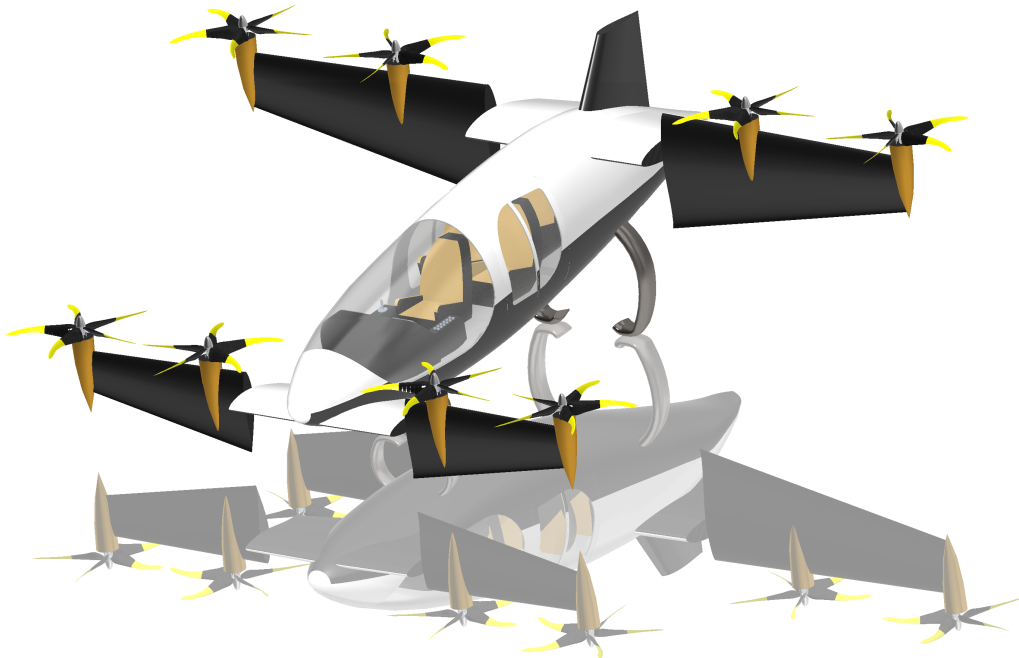
A trade off analysis of the power requirements for selection of the baseline configuration of an electric on demand personal aircraft concept has been performed in this technical report. The design phase is continued with a detailed analysis of the options for sources of energy for the aircraft. This is followed by the actual design of the selected lithium ion batteries. This design was shifted once the performance results were calculated as the power requirement had been reduced and the energy requirement had increased. The report also outlines the selected propulsion, navigational, HUMS and fly by wire systems design. Lastly, a weight and balance estimation has been presented. To meet the mission profile after the first iteration, cruise speed and fuselage design had to be changed. The current design does meet the required parameters. However, some problems have been identified and in further iterations they should be fixed.

Acknowledgments

The author would like to acknowledge the help received from all supervisors during the duration of the project. The team also needs to be acknowledged for their outstanding technical contributions and the great group environment.

Colaboration

This was a highly collaborative project. During the first stages of the design, all teams worked very closely together to decide on the initial configuration. During initial sizing, the Systems and Propulsion team worked in collaboration with the Aerodynamics team to improve the initial sizing script. In later stages of the design, the author collaborated together with the Performance Analysis team. During the whole duration of the project close collaboration with all the members of the Systems and Propulsion team was carried out. Most of the work presented regarding weight and balance, batteries and systems was done together with Y. Bardi Vale [1]. It is recommended to read first [1] as it has a more extensive technical introduction to batteries. Although redundancies with [1] have been avoided, there is some overlap to maintain the coherence of the reports on their own.



Contents

Introduction	1
1 Motivation	1
2 Constraints and Goals	1
 Baseline Configuration	 1
3 Initial Configuration - Cruise vs Take off efficiency	1
 Batteries	 3
4 Full electric - Hybrid propulsion	3
5 Battery selection	3
5.1 Energy options	3
5.2 Reliable battery information	3
5.3 Hybrid batteries	4
5.4 Implementation into initial sizing	4
5.5 Improving energy usage during take off and landing	4
5.6 Results	5
5.6.1 Final battery design from initial sizing	5
5.6.2 Battery design after performance estimation	5
5.6.3 Final battery assuming a more optimistic improvement	5
6 Battery failure and safety	6
 Systems	 7
7 Propulsion system	7
7.1 Cable weight per metre	7
7.2 Inverter weight per Watt	7
7.3 Propulsion system layout and redundancies	7
8 Navigational sensors & Autopilot	7
9 Lights	8
10 Health and Usage Monitoring System (HUMS)	8
10.1 Battery	8
10.2 Proprotor and hub	8
10.3 Motors	9
10.4 Airframe & tilting mechanism	9
11 Control surfaces	9
12 Fly by wire	9
 Weight and Balance	 9

13 Weight and CG estimation	9
14 Results	11
Conclusion	12
15 Summary	12
16 Further Work	12
References	13
Appendix	16
A Uber elevate 2017 simplified analysis	16
B Original Ragone plot	17
C Updated values after implementing battery selection into initial sizing	17
D Manufacturers cable sizing tables	18
E Battery failure diagram	19
F HUMS system layout	19
G Battery usage	20

List of Figures

1	Disc Loading vs power required. Fixing span and changing number of motors	1
2	L/D vs power required. Fixing span and changing speed	2
3	Optimisation for different mission variants	2
4	Ragone plot. Improvement of specific energy and power	4
5	Average take off and landing speed vs MTOW	5
6	One of the two independent propulsive system layout (batteries are denoted in brown)	7
7	Fly by wire systems layout	9
8	L/D and disc loading vs power for the different concepts [8]	16
9	Realistic values for all the different concepts [8]	16
10	Ragone plot (specific power density in W/kg vs. specific energy density in Wh/kg) [16]	17
11	Manufacturers cable sizing table [27]	18
12	Battery failure diagram [26]	19
13	HUMS system layout	19
14	Battery usage	20

List of Tables

1	Mission characteristics	2
2	Efficiencies	5
3	Summarised values for battery design	6
4	Position and weight of components	10
5	CG position for different loading cases	11
6	Updated values after implementing battery selection into initial sizing	17

AVA Team Members

F.G.	Farri Gaba	Project Co-ordination and CAD
B.U.	Birk Ulstad	Project Co-ordination and CAD
J.H.	Jacob Holford	Aerodynamics
A.L.	Alexandros Lessis	Aerodynamics
K.W.	Kathy Wu	Aerodynamics
S.C.	Shin Chang	Structural Design
K.O.	Kenechukwu Obi	Structural Design
M.R.	Manoj Rajendran	Structural Design
Y.V.	Yago Bardi Vale	Systems and Propulsion
A.M.	Arnaud Mortelette	Systems and Propulsion
J.L.	Jorge van Looy Lopez	Systems and Propulsion
E.B.	Elizabeth Wood Bowyer	Systems and Propulsion
A.A.	Ayanle Aden	Flight Mechanics and Performance
M.M.	Marwan Mousa	Flight Mechanics and Performance
X.T.	Xin Kai Tan	Flight Mechanics and Performance
J.T.	Jonathan Tilford	Flight Mechanics and Performance
L.G.	Loris Gliner	Flight Simulation
B.N.	Benita Nortmann	Flight Simulation

List of Symbols

ρ	Density [Kg/m ³]
D	Drag [N]
DL	Disc Loading [N/m ²]
$HUMS$	Health and Usage Monitoring System
L/D	Lift to Drag ratio
$MTOW$	Maximum Take Off Weight [N]
OEI	One engine inoperative (in this case one motor inoperative)
P	Power [W]
T	Thrust [N]
t	Time [s]
V	Velocity [m/s]
$VTOL$	Vertical Take Off and Landing
W	Weight [N]
AVA	Advanced VTOL Aircraft
BLDC	Brushless DC motors
EVS	Enhanced Vision System
FAA	Federal Aviation Administration
FATT	Functional Area Technical Team
GPS	Global Positioning System
I	Intertia [m ² kg]
INS	Inertial Navegational System
LED	Light-Emitting Diode
Pax	Passangers
PCM	Powertrain Control Modules
PTC	Positive Thermal Coefficient
Specific Energy	Watts hour per kilogram [Wh/kg]
Specific Power	Watts per kilogram [Wh/kg]
SVS	Synthetic Vision System

Introduction

1 Motivation

On demand personal electric aviation has the potential to revolutionise two aspects of how transport between cities is done. Firstly, they provide a zero local emission transportation capability, while helping to move away from hydrocarbon-based approaches. Secondly, it can radically improve urban mobility. The time spent travelling between cities can be reduced by more than half [2] while reducing the cost by 90% when compared to helicopters [3]. The whole aeronautical industry is aware of these capabilities and currently there are more than 80 different models of VTOL aircraft being developed [4].

2 Constraints and Goals

The main objective at the beginning of the project was to design a high performance (Minimum speed of 120 knts for cruise) electric personal aerial vehicle that is able to take off vertically. This vehicle should be able to carry three passengers plus the pilot. To be able to achieve these objectives the Systems and Propulsion team were faced with unfamiliar challenges compared to traditional aircraft design. High performance electric aircraft have been a feasible idea for only a few years as the battery technology was not advanced enough [5]. Therefore, as will be shown further in the report the major challenges faced during the design are mostly related to obtaining enough power and energy from the batteries. On the other hand, electric propulsive system are relatively simpler than combustion propulsive systems, as there are fewer moving parts.

A challenge that horizontal take off electric aircraft such as the Pipistrel Alpha Electro do not have to face is the high power requirement. Normally the energy requirement for a typical mission, is much more constraining than the power requirement and by fulfilling it, the power requirement is automatically fulfilled. However, during vertical take off the amount of power required is extremely high. This had a great influence in the battery selection.

Regarding systems, safety and ease of controllability were set as the main priority and therefore a highly failure proof propulsion system was designed. In the future it is expected that AVA pilots will only have to undergo reduced training, therefore, a highly precise navigation system was designed. In addition to this, a Health and Usage

Monitoring System was implemented to reduce any unnecessary risks and provide the pilot with useful information during flight.

Baseline Configuration

3 Initial Configuration - Cruise vs Take off efficiency

The initial configuration was decided together with all the FATTs using both qualitative and quantitative analysis. One of the main aspects that influenced the final decision was the trade off between hover and cruise efficiency.

As this analysis was done at the beginning of the project a simplified model was used. To calculate how efficient each of the possible designs is at hover, Equation (1) was used, which comes from simplified momentum theory.

$$P = \sqrt{\frac{T^2 DL}{2\rho}} \quad (1)$$

Trust can be substituted by weight as the difference between hover and take off is approximately 4% [6].

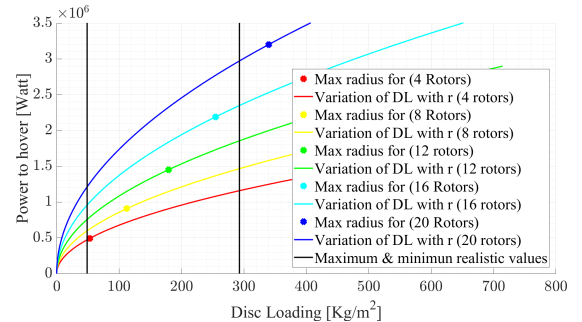


Figure 1: Disc Loading vs power required. Fixing span and changing number of motors

$$P = T * V = \frac{W}{L} * V \quad (2)$$

To obtain Figure 1, two variables had to be fixed: span and clearance between propellers. These limits are reasonable as the span is highly limited in order to reduce the size of the platform and the clearance between propellers will almost not vary depending on the number of motors. A clearance of 0.1m was set between propellers. This implies that for each of the four motors the available span for the blades is reduced by 0.4m (taking into account that there is also clearance with the cockpit). To calculate the efficiency at cruise Equation 2 was used. Ideally, an exact relation between disc

loading and drag should be used. However, this was extremely complicated to model in the first week of the project. Therefore a realistic value approach was followed. This can be done as it is clear that a lower wing loading (when keeping the weight constant) implies a higher disc area and therefore a higher parasitic drag which translates into a lower lift to drag ratio [7]. Thus, the more efficient at hover the less efficient at cruise and vice versa.

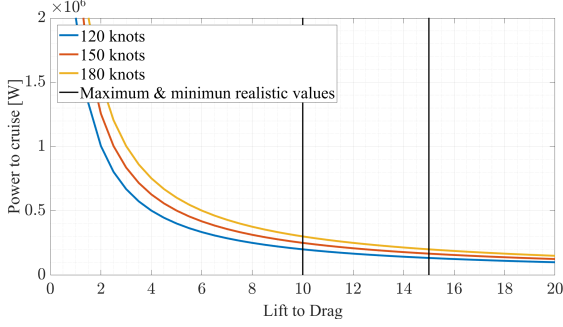


Figure 2: L/D vs power required. Fixing span and changing speed

This method not only provides a clear trade off for the selected configuration but also with a fair study regarding power between other VTOL designs. This study was not coded into the script as the analysis had already been done by [8] in 2017 for the Uber Elevate summit. Their results can be found in Appendix A.

Using the limits of the realistic values provided by [8] allows not only to see that the 20 motor option is too inefficient at hover (Figure 1) to be considered but also to define a range at which the design will be operating for both cruise and hover. This range can be seen in Figure 3. As the mission is defined, it is possible to see what should be given more priority optimisation wise by plotting constant energy lines. This can be obtained as $E = P * t$ and the cruise speed is known to be 120 knts (given by the mission). It also possible to see how sensible this optimisation is by checking different variations of the main mission (Parameters of each mission plotted can be found in Table 1).

Table 1: Mission characteristics

	Typical	More hover	Less distance	More Range
Range (km)	120	120	60	170
Time (s) hovering	200	400	200	200

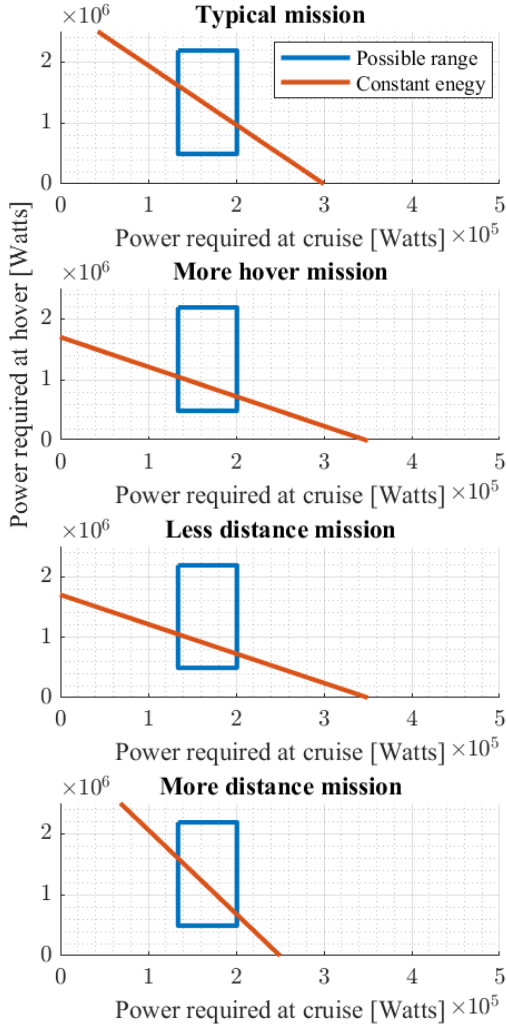


Figure 3: Optimisation for different mission variants

The blue boxes in Figure 3 indicate the realistic range where the design should be. It is not possible to design for the left bottom corner as there is a trade off between both efficiencies. What can be appreciated from all the variants of the main mission is that the top left corner (which indicates the point where cruise has been optimised at the cost of not optimising hover i.e. more but smaller propellers) is always less energy efficient for the mission than the right bottom corner (where the design has been optimised for hover i.e. less but bigger propellers). Therefore, the clear recommendation from this trade off analysis was that the 4 and 8 motors options were clearly more energy efficient than the 12 and 16 motors options. However, there are a number of factors that have not been considered in this analysis that needed to be taken into account when making the final decision, such as safety and controllability. These

are described in reports [9] and [10].

Batteries

Most of the work presented in this section was done in collaboration with Y. Bardi Vale [1].

4 Full electric - Hybrid propulsion

A hybrid concept for AVA was considered at the early stages of the design. The main advantages of a hybrid design is the fact that it would allow the aircraft to reach the 120km desired range without mayor difficulties. It will also provide a percentage of the power required which will allow for a battery system with a much better specific energy. Furthermore, it will also ease the task of releasing AVA by 2021. However, this type of design will produce more noise and pollution and will become outdated relatively quickly when battery technology reaches the desired characteristics (Approximately 2025-2028). As this is a long term project this last disadvantage makes a hybrid system undesirable. A full electric system allows the model to be updated easily with the latest battery technology by just substituting the batteries.

5 Battery selection

As aeronautical engineering students not much was known about batteries before starting this design. An introduction to the key concepts necessary to understand how to address battery selection can be found in [1]

5.1 Energy options

Numerous options were considered to obtain the energy for the aircraft: solar panels, supercapacitors, lithium ion batteries, zinc-air fuel cells, lithium sulphur batteries and lithium metal batteries. The last two options were discarded as they are relatively new types of batteries that will not reach the necessary characteristics by 2021 [11, 12, 13]. Solar panels were also disregarded for many reasons, the main one being the necessity of a clear sky and the high weight penalty. Zinc-air fuel cells almost triple the specific energy of lithium ion batteries such as the ones used by Tesla, however, their specific power is not sufficient. In addition to this, their performance is very sensitive to humidity and they can produce highly flammable hydrogen when

corroded (although they are safer than lithium ion when not corroded as they do not require volatile materials).

Therefore, two options were considered during the main design phase of the battery system: supercapacitors and lithium ion batteries.

5.2 Reliable battery information

It is worth mentioning the huge variance in information regarding the current and future state of batteries. For example, for lithium ion batteries both Reference [8] and Uber in the Elevate summit use a specific energy of 300Wh/kg (To be more precise, Uber Elevate assumes a specific energy for cells of more than 400Wh/kg, assuming a packing efficiency of 1.3, which is the same as the one achieved by Pipistrel Alpha Electro [2], it implies a battery density of around 300Wh/kg). However, in the same summit where both papers were presented the CEO of Pellion Technologies (a battery manufacturer) stated that none of the battery technology being currently used for electric aircraft is close to a specific energy of 250Wh/kg [14]. As an example of realistic values, Airbus E-fan currently uses a battery with an energy density of 150Wh/kg [15]. A similar mismatch between actual characteristics and advertised characteristics is found in supercapacitors.

It is clear that there is going to be a relatively big leap when zinc-air fuel cells, lithium sulphur batteries and lithium metal reach their full potential [11, 12, 13]. However it is almost certain that this technology will not be available by the objective release date of 2021.

Therefore, it was decided to use the data from [16] (specifically Figure 10) which shows realistic characteristics (much smaller than the 300Wh/kg stated by Uber) for batteries. This is corroborated by [17, 18, 19]. As the data is from 2008, two different improvements for 2021 were assumed. Firstly a more pessimistic improvement of 5% per year has been assumed as this has been the average improvement lithium ion batteries in the past 20 years [20]. This is a relative conservative approximation as in the past few years the investment and advancements in battery have increased considerably [21]. The more optimistic improvement assumes a 5% improvement per year until 2015 and a 10% improvement per year from that year onward. This 10% seems to be the trend for the past couple of years [22].

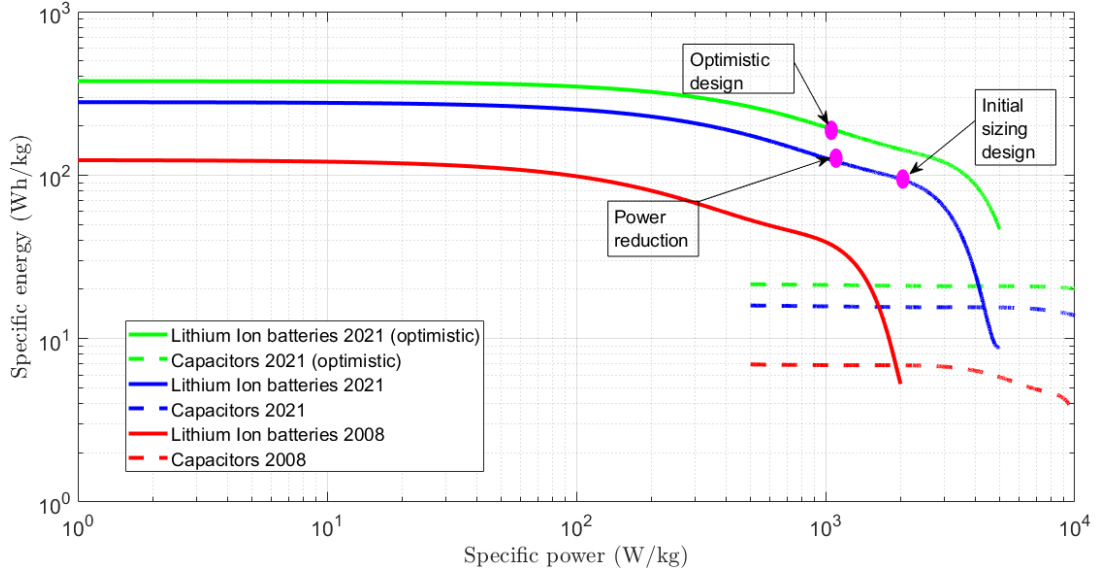


Figure 4: Ragone plot. Improvement of specific energy and power

Another factor that is barely mentioned by both manufactures and clients for VTOL aircraft is the specific power. For example, Uber, assumes that once the energy requirement is fulfilled the power requirement will also be fulfilled.

This is true for most of the horizontal take off electric applications where the power required is much smaller. For VTOL applications which require a high cruise speed such as AVA (and therefore it cannot be optimised only for take off) the power required during take off and landing will be of the order 10⁶ W, assuming an approximate weight of 2 tons. This is much more than what a similar sized helicopter requires as it is optimised for hover. A battery with less than a 1,000W/kg specific power will most probably account for half of the weight of the aircraft. As seen in Figure 4 once specific power goes beyond 1,000W/kg specific energy starts decreasing dramatically. This shows a clear trade off on battery characteristics that was optimised when coded into the initial sizing script.

5.3 Hybrid batteries

To do the battery selection a script was developed that ran over all the possible combinations of lithium ion batteries and capacitors to obtain the minimum battery weight that fulfilled the power and energy requirements. This was done for three different cases. An explanation of this can be found in [1].

5.4 Implementation into initial sizing

Once the script was developed, it was implemented as a function into the initial sizing script coded by [6]. By doing this the results given by initial sizing became much more realistic and exact as initially they were considering a battery with 300 Wh/kg and 1,500 W/h, which as explained in Section 5.2 is completely unachievable for 2021. This caused a 50% increase in our initial weight estimation in week 3 (See Table 6). In addition to this, the battery selection will adapt depending on whether changes affect the power or energy requirements more. For example, the optimal battery for the design in week three was of 169 Wh/kg and 1,008 W/kg. However, due to the modifications in the design the power required increased significantly during week four making the new optimal battery 129Wh/Kg and 1,163W/kg. All the power and energy requirements are calculated by the main initial sizing script following [7]. For this calculation the efficiencies shown in Table 2 were assumed.

5.5 Improving energy usage during take off and landing

The weight obtained using the initial sizing script once the optimal battery code had been implemented provided unpromising results as the battery weight was close to 1500kg (more than 60% of the total weight). One of the main reasons for this was the fact that we were consuming more than 15% of the mission energy during take off and landing. To solve this problem a trade off analysis was carried out on the average speed for take off

and landing.

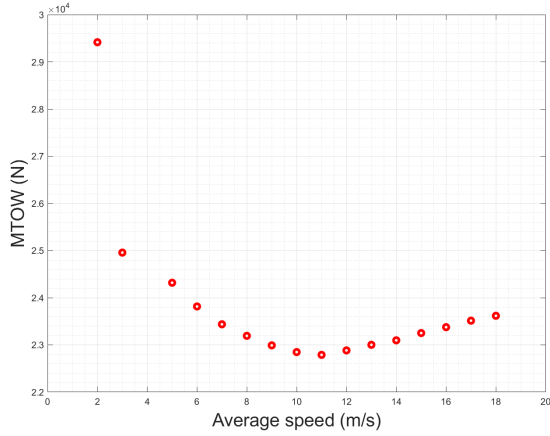


Figure 5: Average take off and landing speed vs MTOW

A higher average speed reduces the time spent for take off and landing, reducing the required energy, thus reducing battery weight. However, it increases the power required and therefore the motor and propotor weight. It can be seen that the optimal speed would be between 10m/s and 12m/s, however this speed is too high to maintain controllability and it would imply an acceleration that would feel uncomfortable for passengers [23]. The acceleration and average speed between take off and landing will differ. Accelerating during the descent is much more unsafe than accelerating during take off, therefore, a slower average speed needs to be used during landing. After discussing it with the Flight Mechanics team, an average speed of 6m/s was selected for landing. For take off, an average speed of 11m/s with an acceleration of 2.7 m/s^2 was selected. This gives a total weighted average speed of 7.5 m/s. Although the power required increases the difference is minimal (From 3m/s average to 7.5 m/s average the power required increases by only 3%). At an average speed of 7.5 m/s the percentage of energy used for take off and landing is reduced to less than 5%.

5.6 Results

Summarised values for battery design for the three design milestones can be found in Table 3. As can be seen battery weight is almost 50% of the aircraft which is very significant.

5.6.1 Final battery design from initial sizing

Using an iterative process a battery that fulfilled both the energy and power requirements was cal-

culated during initial sizing. The first sizing of the battery was done with the pessimistic improvement assumption.

Table 2: Efficiencies

Component	Efficiency initial sizing	Efficiency performance
Proprotors hover	0.80	0.80
Proprotors cruise	0.70	0.74
Motor	0.85	0.95
Inverter	0.98	0.98
Cabling	0.97	0.97
Total hover	0.64	0.71
Total cruise	0.56	0.65

5.6.2 Battery design after performance estimation

Once the performance analysis was done, it was discovered that the initial sizing script was overestimating the power required. Even though weight had increased from the initial sizing estimation (See Section 13) efficiencies had gone up significantly. Therefore, it was possible to shift the battery design (using the data in [4]) to a design with higher specific energy and lower specific power. This was done keeping the mass constant so that no other FATT would be affected. As explained in [24] even with this increase in specific energy the range of 120 km is still not achieved due to an overestimation of L/D in the initial sizing script. In addition to this, an overestimation of lift forced a change in the cruise speed from 120knots to 140 knots.

5.6.3 Final battery assuming a more optimistic improvement

Due to the fact that the performance estimation gave a total range of approximately 77 km [24] the more optimistic improvement (As explained in Section 5.2) was applied to the batteries. It is quite complicated to assure with certainty whether or not the optimistic values presented in Table [3] will be reached by 2021. However, the current interest in EV will definitely push this technology at an unseen pace in the coming years making this optimistic approach realistic.

Table 3: Summarised values for battery design

	Initial sizing	Performance shift pessimistic improvement	Performance shift optimistic improvement
MTOW (kg)	2,353	2,495	2,495
Max power required (kW)	1,218.2	885.0	885.0
Power available after efficiencies (kW)	1,218.2	866.8	866.8
Battery mass (kg)	1,188.8	1,188.8	1,188.8
Specific power (W/kg)	1,163.6	1,024.7	1,024.7
Specific energy (Wh/kg)	129.1	149.0	194.4
Volume (m ³)	0.7	0.7	0.7
Range (km)	125	74	130

The two different improvements show at what range AVA is going to be operating in the first year of operations. As it can be seen, the design is extremely sensitive to battery improvement. Although maximum power required is not met, this value is given by the last seconds of take off were the aircraft reaches 11m/s of upward velocity. By reducing this velocity to just 10m/s all power requirement will be met. The OEI case which gives the *critical* maximum power is met with a safety factor of 10% [24].

6 Battery failure and safety

There are three main sources of battery failure: from wrong use of batteries (for example taken into too high temperatures), from operational accidents (crashes) and from spontaneous internal failures. As AVA will be operated in very restrained conditions by operators (and not individuals) failures due to wrong use of the batteries is not a major concern.

Failures due to spontaneous failures are highly unlikely. It has been estimated to be between 1 in 10 million and 1 in 40 million per cell thanks to the thorough inspection that battery manufacturers have to follow [25].

When looking at the battery cell level most failures normally start propagating in the same way: The temperature of an individual cell starts to increase due to some failure [26]. As decomposition of reactants creates more heat and gases, the “failure zone” expands. From this point there are

three possible outcomes:

1. Heat dissipation rate exceeds heat generation rate: the cell is cooled down and there is no failure.
2. Contained explosion, heat dissipation rate is less than heat generation rate: the heat causes the cell to explode but it does not effect the adjacent cells.
3. Uncontained explosion, heat dissipation rate is less than heat generation rate: the heat causes the cell to explode and it affects the adjacent cell causing a chain reaction.

A diagram of this can be found in Appendix E. Therefore, to avoid this chain reaction a few extra safety characteristics (in addition to the cooling system), have been added to the battery design. A layer of Ceramic PTC material (Positive Thermal Coefficient) is used on cells as it reduces the current automatically when temperature increases. Specifically *ceramic* PTC is used as it returns to its original state once the current returns to normal. Normally, as a last resource, thermal fuses are installed. These will destroy themselves if the current reaches a certain value avoiding a catastrophic failure but forcing a replacement of the thermal fuses. To avoid this kind of replacement it was decided to implement Powertrain Control Modules (PCM), these are attached to the packs as modules. In the event of a failure condition, the PCM opens the battery circuit and prevents damage to the pack. Lastly, vents were also added to the design. These vents open when there is an

increase in pressure in the cell, allowing gases to escape. A safe release of internal pressure avoids dangerous rupture of the cell casing. However, this is also a last resource safety system, as some quantity of the internal composition of the cell is lost when the vent is activated.

Systems

Most of the work presented in this section was done in collaboration with Y. Bardi Vale [1].

7 Propulsion system

One of the many advantages of using an electrical propulsive system is its simplicity when compared to a turbine propulsive system. All the fuel pumps, CG movements due to fuel being burned and fuel dumping mechanisms are not necessary. On the other hand, the cabling will be much heavier and inverters will be necessary.

7.1 Cable weight per metre

To determine the weight of the cables providing the power from the batteries to the inverters and to the motors, manufactures' data sheets were used. As the components are critical a 3% resistance is assumed following the electrical design convention. Using the voltage provided by the batteries [1], and the power required [9] it is possible to calculate the current ($P = IV$). Using the tables in Appendix D, [27] (which require the current to be known) it was determined that the cable for the main propulsion system is going to weight 1.2 kg per metre with a diameter of approximately 10 cm.

7.2 Inverter weight per Watt

The motors selected are BLDC (Brushless DC motors) contrary to what their name indicates BLDC motors need the power to be provided as AC and therefore inverters are necessary (batteries provide power in direct current). In low power circuits the weight of the inverters can be neglected at this stage of the design. However, due to the high power requirements, even with the most advanced inverters the weight will be considerably high. As with most electric motors the inverter weight varies linearly with the power that goes through it. It was found that the lightest scalable inverter available currently for this kind of application was the Siemens NextGen converter, which has a specific power of 20 kW/kg [28].

7.3 Propulsion system layout and redundancies

Due to the proximity during operation to cities and the uncertainty regarding regulation the propulsive system was designed so that if any one of the critical components (batteries, inverters, motors) fails AVA will be able to land safely. As explained by [6] when one motor becomes inoperative, the opposite motor in both the x and y axis will be turned off to maintain equilibrium. To be able to do this with any kind of single failure, two independent systems were designed. One providing power to the left rear and right front motors and the other providing power to right rear and left front motors. Each independent system is composed of two batteries and four inverters and they provide energy to half of the motors. This is shown in Figure 6. For example, if one of the four batteries fails two opposite motors will be turn off so that the remaining battery (which originally was providing 50% of the energy to 4 motors) powers 100% of two motors. Something similar will happen if an inverter or a motor fails. Inverters' weight varies linearly with power, the number of inverters will not increase the weight as the total power that needs to be converted will be the same. The only weight penalty that comes from this redundant system is caused by the cabling. It is calculated that that the total cable length will be 44.07m and therefore the total cable mass will be 52.88kg. Without these necessary redundancies the cable weight could be reduced to less than half. The inverters' weight was calculated to be 60.93kg.

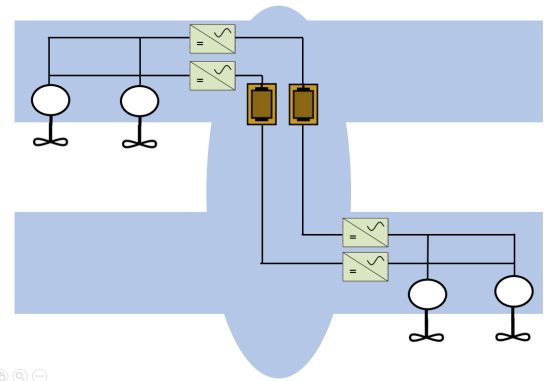


Figure 6: One of the two independent propulsive system layout (batteries are denoted in brown)

8 Navigational sensors & Autopilot

Due to regulation constraints the introduction of the aircraft in 2021 will be done using professional pilots [29] [3]. It is expected that further in the

future AVA will be developed into an aircraft that can be flown with reduced training, similar to the training required to drive a car. The software update required for this is easy to install on the existent avionics computers. On the other hand, including the necessary hardware for a leading autopilot once the aircraft is in operation can cause numerous redesign problems. Therefore, it was decided to include the necessary navigational systems from the beginning, following a similar business plan to the one of Tesla regarding its autopilot software [30].

- GPS/INS: to obtain an accurate position, velocity, attitude and angular rate the MIDG II INS/GPS [31] was selected as it combines Inertial Navigation System (INS) with Global Positioning System (GPS) contained in a small package. It outputs true heading, pitch, roll, position, velocities and accelerations.
- Although the accuracy of the GPS is relatively high, it was considered necessary to add extra sensors to avoid collision with buildings and other vehicles. A Doppler radar was considered for this application, however, it was found that the smaller sized radars are not precise enough to provide a significant advantage. Therefore, a combination of EVS and SVS was selected [32]. Synthetic Vision System uses a database (that is continuously updated) to show recorded terrain characteristics along and in front of the aircraft's path. Enhanced Vision System provides information from infrared cameras to see across fog and darkness, showing a real-time image of what is ahead of the plane. For the SVS the Honeywell SmartView [33] was selected due to its industry leading technology and small size. For the EVS the Max Viz 2300 was chosen due to its low energy, weight and high precision [34]. The EVS cameras will be placed along all sides of the aircraft similar to what is done in cars with autopilots.

Using the highly advanced autopilot unit *VECTOR* [35], the autopilot software can be tested while in operation using the configurable avionics suits (outlined in [1]).

9 Lights

Two different kinds of light systems are required, for passenger safety and for navigational safety:

- FAR 27 states the necessity of appropriate lighting indicating the exit in case of emergency. He-

licopter Emergency Egress Lighting Systems by Cobham [36] were selected due to their simplicity and low weight.

- Collision lights need to be positioned along the aircraft. The Dual Mode Anti-collision lights from Cobham were selected for the fin, for the sides the Cobham LED navigational lights were chosen. Finally for landing, the Dual-mode Search and Landing Light was selected [36]. Using the same manufacturer will reduce the acquisition cost. In addition to this, LED lights have a very low energy consumption and a high reliability.

10 Health and Usage Monitoring System (HUMS)

As AVA is a completely new type of aircraft safety is a key issue for its success. A crash in the early stages of introduction to the market would most probably result in failure from the business point of view. In addition to this, if in the future a full autopilot is implemented the HUMS will be able to improve its effectiveness on critical situations. The core and more direct data collected by the HUMS will be transmitted directly to the pilot and the autopilot unit. The more complex data will be stored and analysed after the flight to adjust: time to next inspection, time to next replacement and ideal performance points. A layout of the HUMS can be found in Appendix F.

10.1 Battery

A Battery-Management System is connected to the battery. This will not only allow for charge and cell protection, it will capture the state of charge, temperature, current, voltage and the state of health (which is a measure of a battery's capability to deliver its specified output). It will also store the battery's history, which can be later analysed to create a more accurate estimate of the number of cycles that a specific battery will be able to do. This will be connected to the main aircraft computer providing the pilot with live information on the state of the battery. In case of any kind of problem such as overheating the pilot will know and will be able to react accordingly.

10.2 Proprotor and hub

By adding a number of strain gauges and accelerometers to the proprotors it is possible to monitor the extension/retraction cycles, hours of operation,

the loads on critical points and vibrations. By inputting the material properties into the aircraft computers, it is possible to code warnings if any of the values are too high.

10.3 Motors

Torque and temperatures of the motors will also be monitored. This not only allows for the creation of similar warnings to the ones mentioned previously, it also provides data to tune the ideal performance point of the motor.

10.4 Airframe & tilting mechanism

Similar sensors are used for the proprotors, providing loads at critical points, cycles and vibrations. In the tilting mechanics the pressure of the oil system will also be controlled.

11 Control surfaces

Due to the low forces that the control surfaces need to maintain [37], it was possible to use electromechanical actuators for both the elevons and the rudder. Electromechanical actuators are much lighter than hydraulically activated actuators, however they are less reliable. As they are not critical for emergency landing [37], this lower reliability was considered acceptable.

12 Fly by wire

To obtain the necessary redundancy, separate data channels from subsystems to the fly controls were designed, similar to what Airbus and Boeing do in their aircraft. A layout of the fly by wire system can be seen in Figure 7 Detailed information on

how the aircraft is controlled can be found in [38, 39, 37].

Weight and Balance

Weight and balance estimation was completed as this is an essential task in preliminary design, especially for the second iteration. Centre of gravity was calculated in the longitudinal (from the nose x) and vertical (from the ground z) position. The aircraft is assumed to be symmetric in the lateral direction. This will be true for all the loading cases thanks to the 1-2-1 seat configuration (if some simple seating rules are implemented for the 1 and 2 passengers cases). This section was done in collaboration with Y. Bardi Vale [1]

13 Weight and CG estimation

There are many known methods that outline a sensible statistical approach to obtain the weight of each aircraft component. These methods are relatively accurate when sizing conventional aircraft. As AVA is a highly unconventional aircraft the validity of each of these methods had to be considered for each component. For this reason three different methods were used: Roskam [40], Raymer [41] and Nicolai [42]. The first two methods were used for all the components. If neither provided a reasonable result or a valid equation for this kind of aircraft Nicolai was used instead. If a weight for a component could be obtained using manufacturers or FATTs data with certainty, the statistical methods were not used.

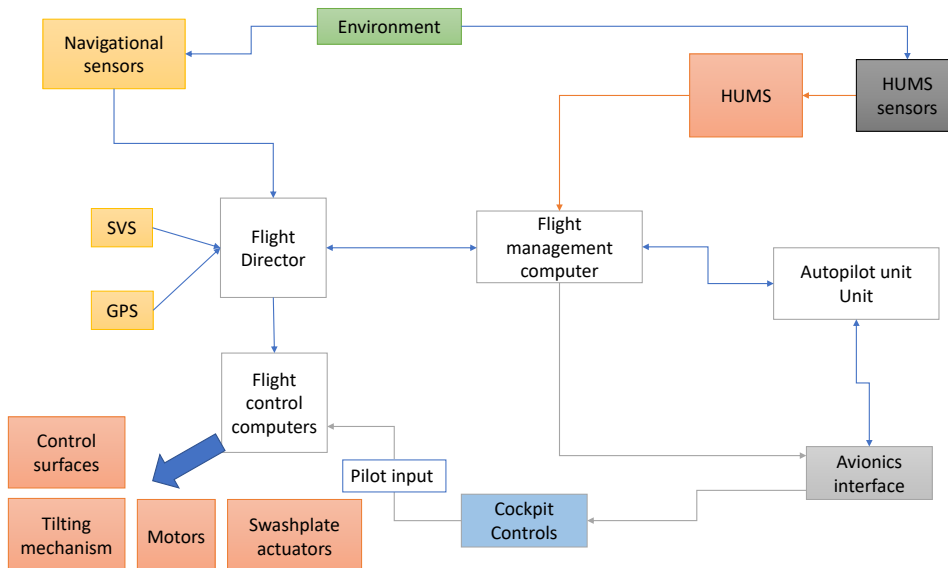


Figure 7: Fly by wire systems layout

Table 4: Position and weight of components

	Roskam (kg)	Raymer (kg)	Nicolai (kg)	FATTs (kg)	Final (kg)	Percentage	x(m)	z(m)
Front Wing	101.78	17.16	73.36	-	73.36	2.94%	0.39	1.00
Rear Wing	127.19	27.44	79.59	-	79.59	3.19%	5.28	2.69
Fuselage	53.55	37.98	-	-	53.55	2.15%	3.55	2.30
Front Landing Gear	-	-	-	81.00	81.00	3.25%	1.85	0.30
Rear Landing Gear	-	-	-	81.00	81.00	3.25%	5.17	0.50
Motors	-	-	-	175.34	175.34	7.03%	2.84	1.85
Proprotors	-	-	-	15.20	15.20	0.61%	2.84	1.90
Proprotors Control	5.12	-	-	-	5.12	0.21%	1.50	1.00
Flights Control	29.57	-	-	-	29.57	1.19%	1.10	1.00
Electromechanical Actuators	-	-	-	37.37	37.37	1.50%	2.84	1.85
Electrical Systems	-	-	-	117.14	117.14	4.69%	2.84	1.45
Avionics	-	-	-	18.14	18.14	0.73%	1.50	1.00
Air Conditioning Unit	-	-	-	16.68	16.68	0.67%	3.50	2.00
Furnishing	72.67	107.51	-	-	72.67	2.91%	3.50	1.60
Parachute	-	-	-	26.50	26.50	1.06%	3.20	2.70
Payload	-	-	-	400.00	400.00	16.03%	2.50	1.50
Batteries	-	-	-	1,188.87	1,188.87	47.64%	3.41	1.72
Vertical Tail Plane	37.22	11.64	3.38	-	3.38	0.14%	6.92	3.1
Tiling Mechanism	-	-	-	20.97	20.97	0.84%	2.84	1.85
Total					2,495.4	100.00%	3.28	1.61

- Wings: the weight of the wing was found to be the most challenging one. It was not possible to obtain a reasonable value from the structures team due to its complexity and use of composites. Roskam's statistical equation does not have as a variable the amount of fuel carried out by the wing (it just assumes that it will carry an amount proportional to the the MTOW) and therefore, provides a value that is considered to be too high. On the other hand, Raymer does have fuel as variable but it does not have into

account the motor weight that the wing will have to support and therefore provides an extremely low number. Although Nicolai does not have into account the fuel carried it provided a much more reasonable value that sits between what was obtained by Roskam and Raymer.

- Fuselage: Both gave relatively similar results. However, one of the variables in Raymer for fuselage is the M.A.C. As it is not very clear how to account for the fact that our design has

tandem wings, Roskam was used.

- Landing Gear: the highly unconventional non retractable carbon fibre landing gear made it impossible to justify the use of a statistical method. Therefore a weight estimation provided by [43] was used.
- Motors: due to their ease of scalability a regression model is used to calculate the weight of the motors [9].
- Proprotors: the weight of the proprotors was calculated by [10].
- Proprotors Controls: Roskam was used as Raymer includes this weight in the propeller weight.
- Flight control: Roskam was used as Raymers includes these in electronics.
- Tilting mechanisms: the weight was calculated roughly when designing the mechanism [1].
- Electromechanical actuators: standard small aircraft electromechanical actuators were assumed for control surfaces [44].
- Avionics: the weight for the G-1000 which is used for aircraft of similar weight [45] was used.
- Air conditioning: the same unit weight as that used in the Cirrus SR-20 [46] was used.
- Furnishing: Roskam was selected as it uses the number of passenger and not the MTOW like Raymer. It has been considered that using MTOW in the statistical models will underestimate the weight, as fuel aircraft with the same number of passengers normally have a much higher MTOW.
- Parachute: a similar weight to the one used in Cessna 172 was used [47].
- Passengers: 100 kg was assumed per passenger as recommended by the FAA [40]
- Fin: a similar approach was followed as with the wing. Although in this case both Raymer and Roskam provided values that were considered too high. In addition to this, it was considered more accurate to use Nicolai to maintain consistency.

The resulting mass of 2,494.3 Kg is an increase of 5.97% compared to the original initial sizing weight

of 2,353.8 Kg [6]. This change may cause a further weight increase in future iterations due to the fact that battery weight may have to increase to fulfil power and energy requirements.

14 Results

To obtain the centre of gravity and inertias the equations below were used.

$$x_{cg} = \frac{\sum W_i x_i}{\sum m_i} \quad z_{cg} = \frac{\sum W_i z_i}{\sum m_i} \quad (3)$$

$$\begin{aligned} I_{xx} &= \sum m_i [(y_i - y_{CG})^2 + (z_i - z_{CG})^2] \\ I_{yy} &= \sum m_i [(x_i - x_{CG})^2 + (z_i - z_{CG})^2] \\ I_{zz} &= \sum m_i [(x_i - x_{CG})^2 + (y_i - y_{CG})^2] \\ I_{xz} &= \sum m_i (x_i - x_{CG})(y_i - y_{CG}) \end{aligned} \quad (4)$$

I_{xy} and I_{yz} are equal to 0 due to the symmetry of AVA.

As no fuel is being burnt during the flight the CG position will only vary depending on the number of passengers. Table 5 summarises each of the main cases.

Table 5: CG position for different loading cases

	X cg (m)	Z cg (m)
0 pax - 0 pilot	3.24	1.63
0 pax - 1 pilot	3.21	1.63
1 pax - 1 pilot	3.26	1.63
2 pax - 1 pilot	3.23	1.62
3 pax - 1 pilot	3.28	1.61

The difference between the most afterwards and forward CG position is 7 cm, an extremely low variation compared to fuel powered aircraft. For this reason a plot with the C.G. variation was considered unnecessary.

The inertias were found to be the following:

$$\begin{aligned} I_{xx} &= 2,622 m^2 kg & I_{yy} &= 6,367 m^2 kg \\ I_{zz} &= 6,779 m^2 kg & I_{xz} &= 908 m^2 kg \end{aligned} \quad (5)$$

However, this is an approximation using point masses and parallel axis theorem. In future iterations a better estimation should be obtained using the CAD model. It was considered unnecessary to present the inertias for the other loading cases due to the roughness of the approximation and

the small changes in weight compared to fuelled aircraft.

Conclusion

15 Summary

After six weeks of design the team has concluded that AVA is a completely feasible aircraft that can be built in the coming years, however, its range will be highly dependent on the battery technology and the advancements in the industry. As it is shown in Table 3, the range of AVA will vary between 74 km and 130 km depending on how much the battery technology will improve by 2021. To be able to meet the mission requirements a final redesign of the fuselage and an increase of the cruise speed had to be done in the last week of the project and this should be improved in the future.

To meet the release date requirements a lithium ion battery was selected. During the analysis it was concluded that a hybrid of batteries would not produce a weight reduction, therefore, only one type of battery was chosen. From the literature review, it was clear that predicting the exact state of the lithium ion batteries in 2021 is not possible and therefore two estimates were used for comparison purposes.

As safety is one of the key parameters for the success of AVA in its early stages, it was paramount to design redundant and reliable systems. For this reason, a propulsion system which can support any single failure of one of the critical components was designed. In addition to this, HUMS was included in the system designed to provide the pilot and the operators with more reliable information that will minimise any kind of failure risk. The batteries have been designed with three extra safety mechanisms that make a chain reaction failure extremely unlikely.

Regarding the navigational system, although AVA will be introduced to the market using professional pilots, its advanced navigational hardware will allow for an implementation of an autonomous

autopilot in the future.

A weight and balance estimation is presented, with results that differ 6% from the initial sizing weight. CG estimations show that the lack of fuel makes the CG variation minimal with a variation of less than 8 cm in the most extreme cases.

16 Further Work

In future iterations of the design there are a number of general issues that need to be addressed, such as the fuselage design and orientation, and the cruise speed required. Regarding the work presented in this report, in further iterations a more precise battery performance could be obtained by obtaining cell data from manufacturers or even testing cells in an experimental setting, to be able to do the modelling of the batteries using our own data.

System wise, a more detailed design should be done regarding flight computers and its redundancies. For example all redundant computers in Airbus aircraft are dissimilar (different chip and processor). Also data busses should be taken into account in future iterations.

Regarding weight and balance, ideally a more accurate value should be obtained for the wings and the fuselage as the accuracy of the current estimates is questionable. The inertias should be obtained using the CAD model. Lastly, if more time was available, penalty weight factors, for the equations used in Roskam, Raymer & Nicolai, could have been developed. This can be done by studying the difference in weight components from aircraft similar to AVA and aircraft used to develop the statistical models of [40, 41, 42]. This will also require detailed information from other VTOL models which currently is not available.

Overall the design of AVA has been an interesting and challenging project allowing us to acquire a great amount of technical knowledge.

References

- [1] Bardi Vale Y. Advanced VTOL Aircraft - Systems and Propulsion. 2018.
- [2] Uber. Uber elevate: Fast-forwarding to a future of on-demand urban air transportation. <https://www.uber.com/elevate.pdf>, May 2017. [Accessed 06/10/2018].
- [3] Ulstad B. Advanced VTOL Aircraft - Project Coordination CAD. 2018.
- [4] Vertical Flight Society. Electric vertical take-off and landing (eVTOL) concepts. <http://evtol.news/>.
- [5] Electric Vehicles: Trends - Mobility & Motors - Pictures of the Future. <https://www.siemens.com/innovation/en/home/pictures-of-the-future/mobility-and-motors/electric-vehicles-trends.html>. [Accessed 06/10/2018].
- [6] Holford J. Advanced VTOL Aircraft - Aerodynamics. 2018.
- [7] Gordon J Leishman. *Principles of helicopter aerodynamics*. Cambridge university press, 2006.
- [8] Rob McDonald Brian German. eVTOL energy needs for uber elevate.
- [9] Wood Bowyer E. Advanced VTOL Aircraft - Systems and Propulsion. 2018.
- [10] Mortelette A. Advanced VTOL Aircraft - Systems and Propulsion. 2018.
- [11] Andreas F Hofmann, David N Fronczek, and Journal of Power Sources Bessler, Wolfgang G. Mechanistic modeling of polysulfide shuttle and capacity loss in lithium - sulfur batteries. 259:300–310, 2014.
- [12] Abbas Fotouhi, Daniel J Auger, Laura O'Neill, Tom Cleaver, and Sylwia. Journal Energies Walus. Lithium-sulfur battery technology readiness and applications—a review. 10(12):1937, 2017.
- [13] Paul Albertus, Susan Babinec, Scott Litzelman, and Aron. Journal Nature Energy Newman. Status and challenges in enabling the lithium metal electrode for high-energy and low-cost rechargeable batteries. page 1, 2017.
- [14] David Eaglesham. Uber Elevate Day 2 Session 1. <https://youtu.be/wpG6XcBNcW8?t=2h10m42s>. [Accessed 06/10/2018].
- [15] Emmanuel Joubert, Denis Chapuis, Didier Esteyne, Jean-Christophe Lambert, Olivier Siri, and Detlef Müller-Wiesner. The E-Fan All Electrical Aircraft Demonstrator and Its Industrialization. 2016.
- [16] Venkat. AIP conference proceedings Srinivasan. Batteries for vehicular applications.
- [17] Patrice Simon and Yury. Jurnal Nature Materials Gogotsi. Materials for electrochemical capacitors. 7(11):845, 2008.
- [18] Maher F El-Kady, Melanie Ihns, Mengping Li, Jee Youn Hwang, Mir F Mousavi, Lindsay Chaney, Andrew T Lech, and Richard B. Journal Proceedings of the National Academy of Sciences Kaner. Engineering three-dimensional hybrid supercapacitors and microsupercapacitors for high-performance integrated energy storage. 112(14):4233–4238, 2015.
- [19] NPC. NPC reports, Electric, chapter 13. http://www.npc.org/reports/FTF-report-080112/Chapter_13-Electric.pdf. [Accessed 06/10/2018].
- [20] Chen-Xi Zu and Hong Li. Thermodynamic analysis on energy densities of batteries. *Energy & Environmental Science*, 4(8):2614–2624, 2011.

- [21] Lithium-ion battery costs & market. <https://data.bloomberglp.com/bnef/sites/14/2017/07/BNEF-Lithium-ion-battery-costs-and-market.pdf>. [Accessed 06/16/2018].
- [22] George Crabtree, Elizabeth Kocs, and Lynn Trahey. The energy-storage frontier: Lithium-ion batteries and beyond. *MRS Bulletin*, 40(12):1067–1078, 2015.
- [23] Armchair Warriors. Engineering the space age: A rocket scientist remembers. by robert v. *Air Power History*, 2010.
- [24] Mousa M. Advanced VTOL Aircraft - Flight Mechanics and Performance. 2018.
- [25] Daniel H Doughty. Vehicle battery safety roadmap guidance. Technical report, National Renewable Energy Laboratory (NREL), Golden, CO., 2012.
- [26] Ph.D Daniel H. Doughty. Failure mechanisms of li-ion batteries. https://www.nts.gov/news/events/Documents/2013_Lithium_Batteries_FRM-Panel1d-Doughty.pdf. [Accessed 06/14/2018].
- [27] Enerdrive independent power solutions. <http://www.enerdrive.com.au/>. [Accessed 06/13/2018].
- [28] Siemens. Electric propulsion components with high power densities for aviation. <https://nari.arc.nasa.gov/sites/default/files/attachments/Korbinian-TVFW-Aug2015.pdf>. [Accessed 06/12/2018].
- [29] Gaba F. Advanced VTOL Aircraft - Project Coordination CAD. 2018.
- [30] Tesla’s big bet: \$8,000 worth of self-driving hardware in all new cars — before the software is ready - vox. <https://www.vox.com/new-money/2016/10/19/13341108/tesla-self-driving-capability>. [Accessed 06/13/2018].
- [31] Cruxial Microbotics. http://www.cruxial.com.sg/mb_gps.htm. [Accessed 06/14/2018].
- [32] Better Together: EVS and SVS - Avionics. <http://www.aviationtoday.com/2014/01/01/better-together-evs-and-svs/>. [Accessed 06/14/2018].
- [33] Smartview synthetic vision system. <https://aerospace.honeywell.com/en/products/safety-and-connectivity/smartview-synthetic-vision-system>. [Accessed 06/14/2018].
- [34] Dual Sensor EVS. <https://www.astronics.com/product?productgroup=Enhanced%20Vision%20Systems&subproduct=dual%20sensor%20evs>. [Accessed 06/14/2018].
- [35] Vector — UAV Navigation. <https://www.uavnavigation.com/products/autopilots/vector>. [Accessed 06/15/2018].
- [36] Cobham plc, aircraft lights, external aircraft lighting. <http://www.cobham.com/communications-and-connectivity/aerospace-communications/aircraft-lights/external-aircraft-lighting/>. [Accessed 06/15/2018].
- [37] Tilford J. Advanced VTOL Aircraft - Flight Mechanics and Performance. 2018.
- [38] Gliner L. Advanced VTOL Aircraft - Flight Simulation. 2018.
- [39] Nortmann B. Advanced VTOL Aircraft - Flight Simulation. 2018.
- [40] Jan Roskam. *Airplane Design: Part V-Component Weight Estimation*. DARcorporation, 1985.
- [41] Daniel P Raymer. *Aircraft Design: A Conceptual Approach (AIAA Education)*. Virginia: American Institute of Aeronautics & Astronautics, 2006.
- [42] Leland Malcolm Nicolai, Grant Carichner, and Leland Malcolm Nicolai. Fundamentals of aircraft and airship design. 2010.

- [43] Chang S. Advanced VTOL Aircraft - Structural Design. 2018.
- [44] Electromechanical flight control actuators — parker na. <http://ph.parker.com/us/en/electromechanical-flight-control-actuators>. [Accessed 06/14/2018].
- [45] Garmin. G1000 — Avionics — Garmin. <https://buy.garmin.com/en-GB/GB/p/6420>. [Accessed 06/12/2018].
- [46] Maintenance & modifications – goodrich aviation. <http://www.goodrichaviation.com/cessna-service-station/>. [Accessed 06/12/2018].
- [47] Cessna 172/182 parachute system — brs aerospace. <https://brsaerospace.com/cessna/>. [Accessed 06/14/2018].

A Uber elevate 2017 simplified analysis

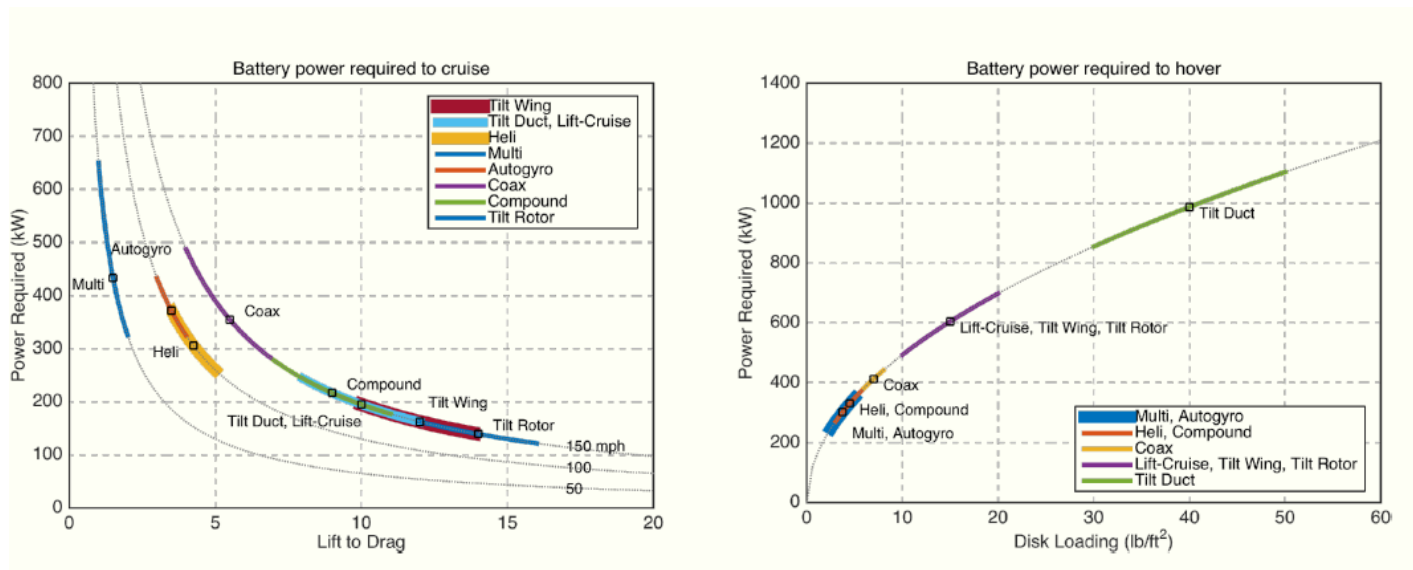


Figure 8: L/D and disc loading vs power for the different concepts [8]

Energy Required For Typical Mission

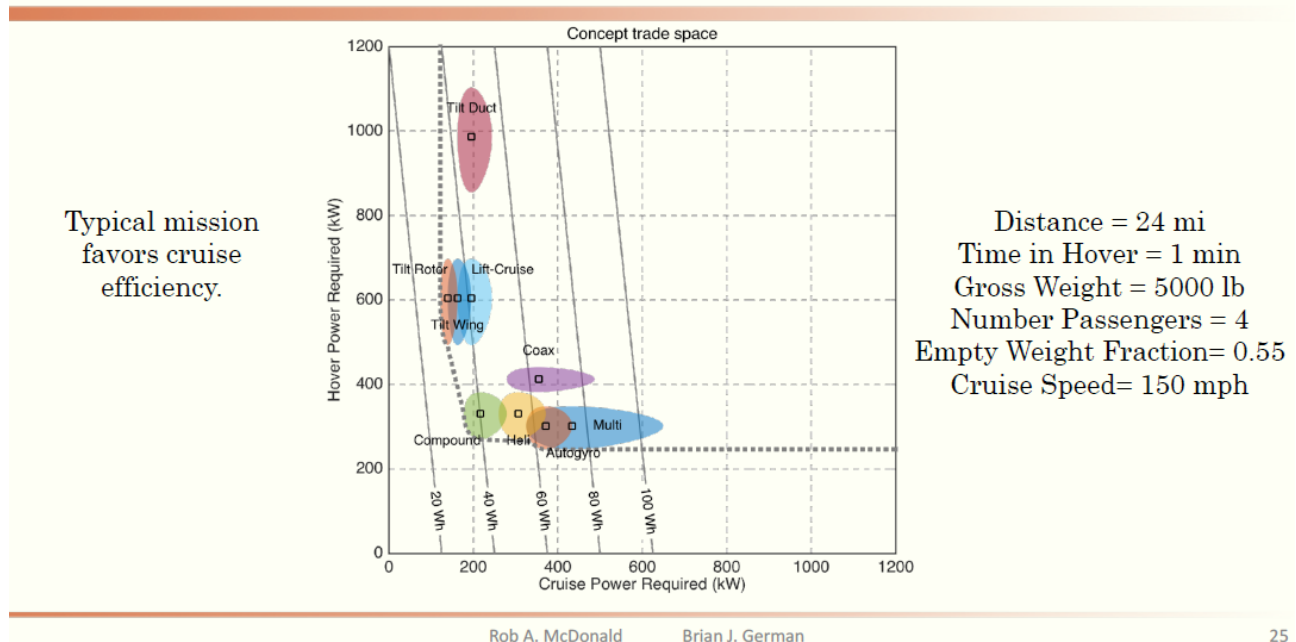


Figure 9: Realistic values for all the different concepts [8]

B Original Ragone plot

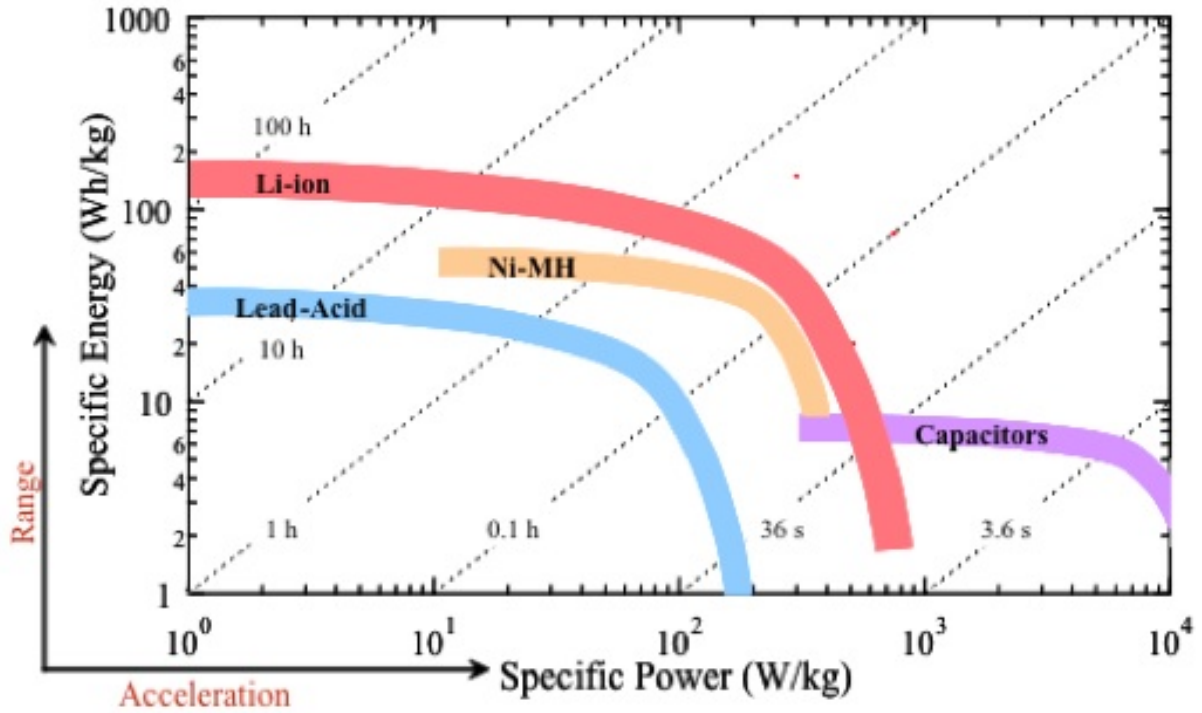


Figure 10: Ragone plot (specific power density in W/kg vs. specific energy density in Wh/kg) [16]

C Updated values after implementing battery selection into initial sizing

These updated values show the jump once the battery sizing function was added. The weight of 2782 was reduced later on by improving the accuracy of other component weights.

Table 6: Updated values after implementing battery selection into initial sizing

	Previous values	Current values
MTOW [kg]	1935	2782
W/S [N/m ²]	1619	1869
W/Sdisc[N/m ²]	2181	1226
T/Wmax	1.275	1.333
Propeller radius [m]	0.588	1.0891
Wingspan (each set of wings) [m]	5.93	Front 6.83 Back 8.37

D Manufacturers cable sizing tables

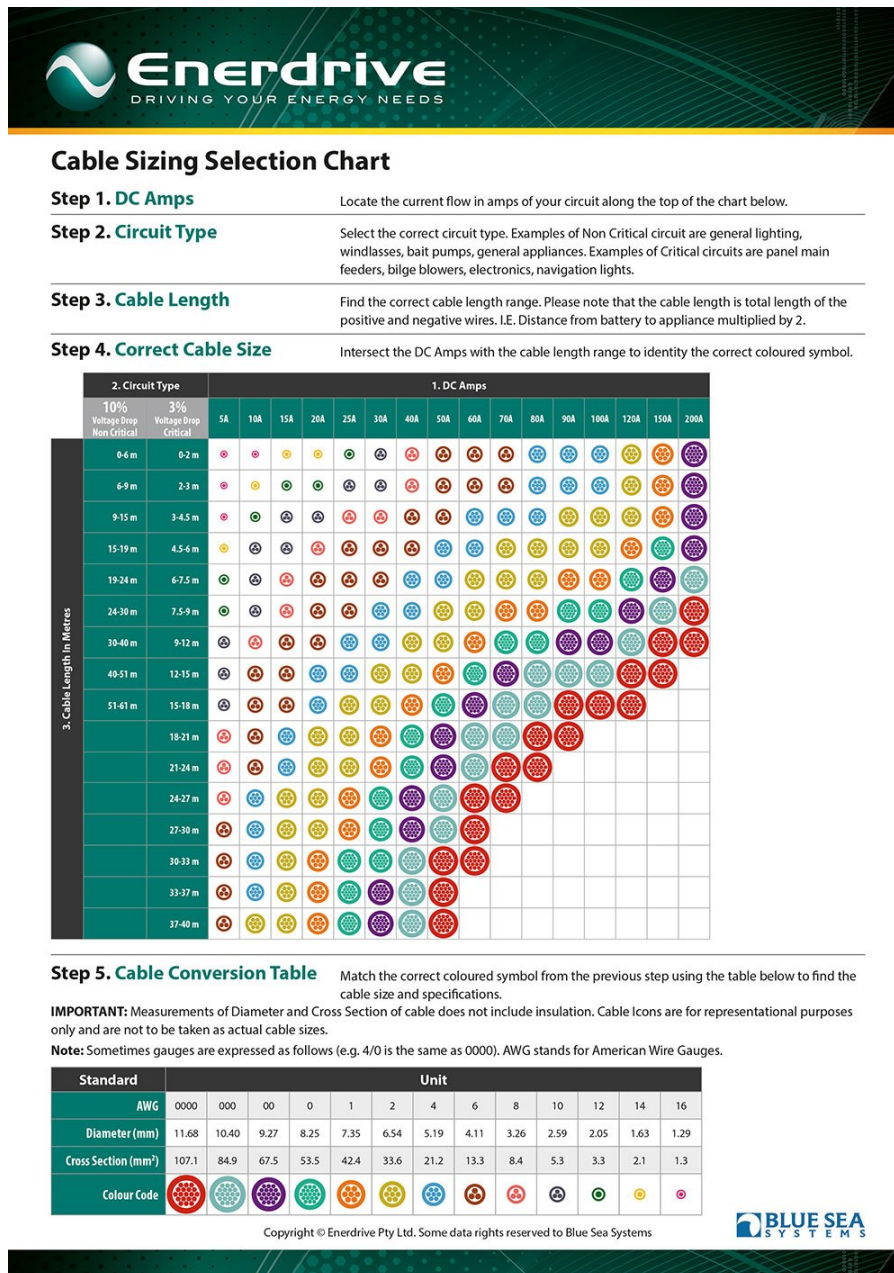


Figure 11: Manufacturers cable sizing table [27]

E Battery failure diagram

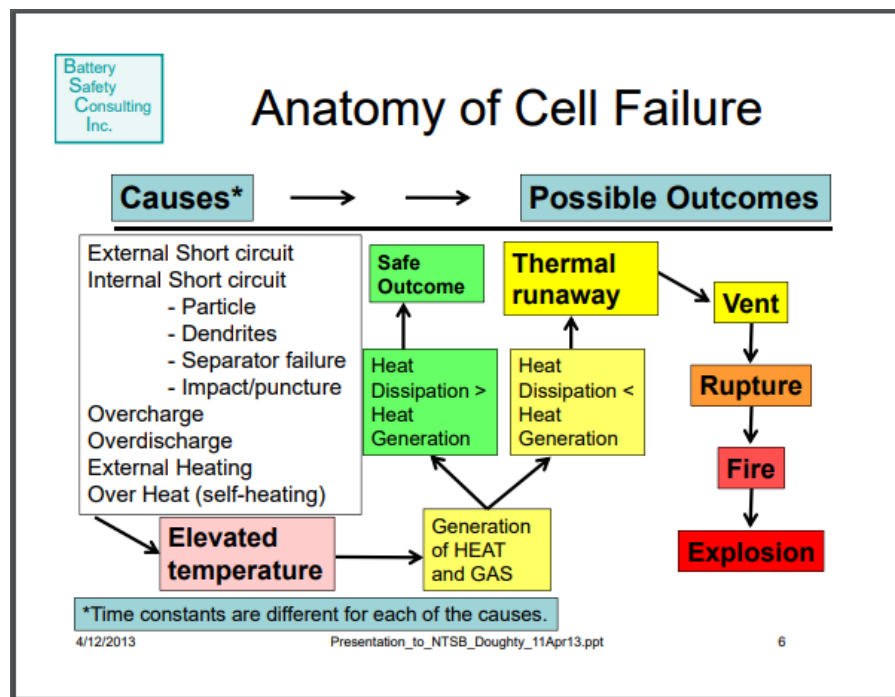


Figure 12: Battery failure diagram [26]

F HUMS system layout

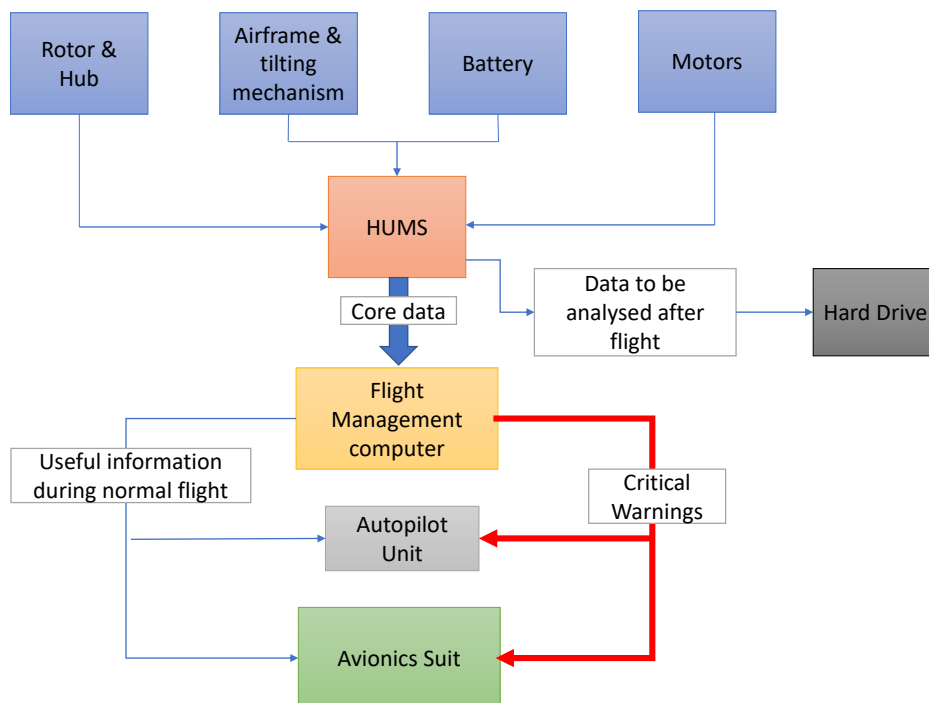


Figure 13: HUMS system layout

G Battery usage

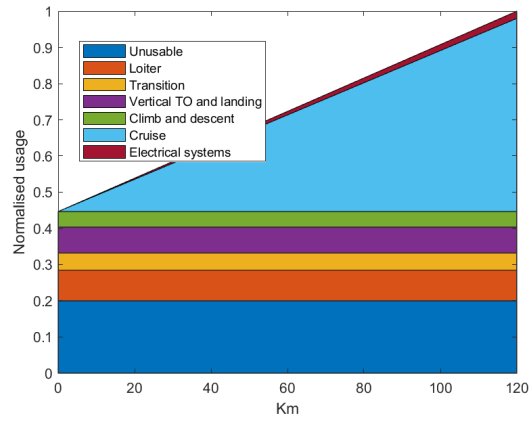


Figure 14: Battery usage

As mentioned more extensively in [1] usage of the battery can be divided into different sections. As can be seen in Figure 14 in a normal mission approximately only 67% of the energy will be used. The rest is saved for emergency situations and to extend the life of the battery.